Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/19/2012
PI Name:	Cooper, Maya M.S.		
Project Title:	Packaged Food Mass Reduction Technology Development		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/25/2012	**End Date:**	11/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date change to 11/30/2012 per PI change in January 2013 to Monica Leong per G. Douglas/JSC (Ed., 2/27/2013)		
Key Personnel Changes/Previous PI:	NOTE: PI change in December 2012 from Maya Cooper to Monica Leong; see Leong for subsequent reports (Ed., 2/27/2013)		
COI Name (Institution):	Leong, Monica (Lockheed Martin)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Advanced Food Technology (AFT) project seeks to define a safe, palatable, and nutritious food system to support long duration space missions beyond Low Earth Orbit (LEO). Balancing resources such as water, food preparation time, stowage volume, launch mass, and power requirements is central to defining a functioning food system solution for these missions. Prior AFT work has addressed the need to decrease the mass of the food system, and has recommended means such as reducing the moisture of the food system, increasing the nutrient density of the food system, and streamlining food packaging systems to accomplish this. Building on these recommendations, the objective of this project is therefore to identify technologies required to formulate meal replacement options that would decrease the mass and volume of the food system, as well as decrease the crew time associated with meal preparation. As commercially available meal replacement products are not adequate for NASA needs, this project will proceed by formulating new products that meet the unique requirements of spaceflight beyond LEO. The development of meal replacement options for the packaged space food system will be achieved by: * Aligning previous studies on nutrition and mass savings to compile the nutritional requirements and consumption guidelines for meal replacement beverages and bars. * Developing meal replacement bar(s) meeting nutrition, sensory, and shelf life requirements for space food systems. * Developing meal replacement beverage(s) meeting nutrition, sensory, and shelf life requirements for space food systems. * Assessing the meal replacement system and creating an implementation plan for the meal replacement options. Some accommodations have been made in the product development design to ensure that the project is completed within budget and on schedule. The recipe development of the bars and beverages will baseline from existing commercial bars and beverages rather than starting from new formulas even though significant changes are likely required to meet nutritional needs. Researchers will leverage flavor concepts from existing commercial bar and beverage products. Accelerated shelf life testing will done to assess stability as the project duration is shorter than the desired shelf life. While the ideal sensory evaluation would be a repeated exposure test to ensure acceptability levels were sustained over multiple experiences, the product amounts and the availability of panelists requires a one-time sensory evaluation for each product. Upon the completion of this study, an assessment of representative formulated meal replacement options will be provided to stakeholders, in order to offer an understanding of meal replacement technology and associated mass savings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The meal replacement technology developed during this task has applicability to the military and disaster relief agencies - two organizations which use heavy amounts of shelf stable goods and are required the transport the goods over large distances. The development of nutritious meal replacement bars potentially alleviates significant dollars in logistical expenses by providing a lower mass, lower volume pathway to supplying adequate nutrition to the desired populations.
Task Progress:	The task work began by aligning previous studies on nutrition and mass savings to compile the nutritional requirements and consumption guidelines for meal replacement beverages and bars. Based on a standard 10-day menu, breakfast meal replacements must deliver approximately over 700 kilocalories and weigh less than 304 grams. Lunch meal replacements must deliver just over 900 kilocalories and weigh less than 610 grams. Requirements for protein, carbohydrates, fat, and dietary fiber were also established for the meal replacements. Secondarily, the following food categories were proposed for mass reduction - dessert and fruit, soups, pudding and yogurt, and a red meat entrée replacement. The first three items were identified in the AFT Nutrition task as having lower than average nutrient density and/or caloric density. The red meat replacement meets the nutritional needs of crew members who desire to reduce their red meat intake while on spaceflight missions. The project successes thus far include establishing the nutritional requirements for meal replacement foods and category foods and the partial completion of the first iteration of bar prototype development. Two bar prototypes have been developed and tested and a third bar prototype is still in an early stage of formulation. Beverage development has also begun, with a focus on applicable ingredient technology and the assessment of solubility of protein powders. The research was executed post-Science Management Office (SMO) Authority to Proceed (ATP), granted on May 25, 2012. Acceleration of research efforts in future months will allow completion of the task as scheduled. NOTE: PI change in December 2012 to Monica Leong; see Leong for subsequent reports (Ed., 2/27/2013)
Bibliography Type:	Description: (Last Updated: 04/23/2019)